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Preventing food waste in subsidy-based university dining systems: An artificial neural network-aided model under uncertainty

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Abstract

Food waste planning at universities is often a complex matter due to the large volume of food and variety of services. A major portion of university food waste arises from dining systems including meal booking and distribution. Although dining systems have a significant role in generating food wastes, few studies have designed prediction models that could control such wastes based on reservation data and behavior of students at meal delivery times. To fill this gap, analyzing meal booking systems at universities, the present study proposed a new model based on machine learning to reduce the food waste generated at major universities that provide food subsidies. Students' reservation and their presence or absence at the dining hall (show/no-show rate) at mealtime were incorporated in data analysis. Given the complexity of the relationship between the attributes and the uncertainty observed in user behavior, a model was designed to analyze definite and random components of demand. An artificial neural network-based model designed for demand prediction provided a two-step prediction approach to dealing with uncertainty in actual demand. In order to estimate the lowest total cost based on the cost of waste and the shortage penalty cost, an uncertainty-based analysis was conducted at the final step of the research. This study formed a framework that could reduce the food waste volume by up to 79% and control the penalty and waste cost in the case study. The model was investigated with cost analysis and the results proved its efficiency in reducing total cost.

Keywords

Food waste management, meal booking, university services, artificial neural networks, error compensation

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Introduction

One of the most challenging aspects of every food supply and distribution system is the minimization of food waste, the success of which ensures the stability of natural systems and management of the growing populations (Silvennoinen et al., 2019). Boosting food waste management has an impact on sustainability in food production and consumption (Martin-Rios et al., 2018). According to a conventional definition, food waste is any surplus or discarded products derived from the process of production, processing, distribution, or consumption of food (Okazaki et al., 2008). According to the material flow model in food services as proposed by Pirani and Arafat (2016), a considerable portion of waste, as well as disposing of food during the preparation and after serving food, is a result of unconsumed food. This type of waste is generally categorized into the post-consumption food waste (PCFW) group, which is hard to manage due to the preparation of final products and the presence of perishable food compounds (Awasthi et al., 2018). Undoubtedly, an accurate prediction of demand and possible food waste can assist food service providers with the right amount of ingredients and in cooking enough meals. On the one hand, overproduction can lead to wasting of food and squandering of financial resources. On the other hand, underproduction may cause some shortage that affects customer satisfaction or food service users (Dolnicar and Grabler, 2004; Lasek et al., 2016). A common approach to managing these issues in food services is demand planning based on order reservations (Ongkunaruk and Kessuvan, 2013). Using the booking systems to pre-order and book meals is an appropriate method for demand planning in food services that can efficiently help reduce overproduction and food waste (Steen et al., 2018). Meal booking systems can significantly reduce food waste by

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estimating the number of daily meals according to the reservation orders (Painter et al., 2016). In order to design booking models that can generate these estimates, a sustainable mechanism for predicting demand is required. Here, the significance of demand prediction becomes obvious. According to Obaidat and Nicopolitidis (2016), future demand prediction is one of the aspects of intelligent systems that can lead to a reduction in costs and an increase in the quality of strategy and business performances. Therefore, managers are required to predict demand to make proper decisions.

The significance of university food waste comes to the fore since it constitutes a significant part of dormitory and campus waste. Furthermore, due to the centrality of food distribution centers at universities, the effectiveness of intervention and planning to reduce food waste will remarkably increase (Abdelaal et al., 2019; Gallardo et al., 2016).

The literature on waste management at universities is mainly focused on analyzing the volume and composition of waste, the causes and factors of waste generation, and results of implementing recycling programs and waste reduction (Gallardo et al., 2016). In a research study conducted at a university in Nigeria, the production of solid waste was studied to estimate the ratio of different waste components. The largest share of waste belongs to food waste, weighing nearly 140 kg per day. The cafeterias and restaurants at this university contributed the most to the production of food waste (Okeniyi and Anwan, 2012). In this regard, another study considered a university in Iran as a case study and investigated solid waste. It was found that the largest daily production of compostable wastes was derived from organic waste. Due to the surplus volume of food distributed daily to students, food waste, that is, the remains of the food distributed in student dormitories or university restaurants, constitute the largest portion of organic waste (Taghizadeh et al., 2012). The production of waste and its compounds was also studied at a Spanish university during 1 academic year. This university uses a university waste management system with two axes: waste management and cleaning service. In the case of university restaurants, the highest proportion of their waste is comprised of organic materials that constitute approximately 61% to 72% of all wastes (Gallardo et al., 2016). In 2016, a study focused specifically on food waste at a university in South Africa. It aimed to estimate the amount of food waste production at the dining halls of universities. Different factors including distance to the dining hall, mealtime, and its ingredients are known to affect the production of food waste, thus prompting each student to produce an average of 0.555 kg of food waste per day. This study further discussed the reasons for the slow rate of students at the dining halls and the diversity of waste productions in different self-services at this university (Painter et al., 2016). A similar study investigated the condition of food waste on a university campus in Qatar. In this research, in addition to revealing a general estimate of food waste in different places, the amount of waste specific to each type of food was presented. One of the key factors involved in the results of this study was overproduction with emphasis on different places for preparing and distributing food on campus (Abdelaal et al., 2019). Besides those studies that have focused on estimating the waste volume, other authors have also examined operational methods for and approaches to reducing waste volume. A study reviewed a university that serves nearly a thousand meals a day and explored the effect of different food delivery scenarios on the amount of waste produced at the university dining hall. This study has investigated the shift from the tray-based system of food delivery to a non-tray-based system in which food waste and customers' perspectives are evaluated, thus leading to reduced solid food waste and high customer support at the buffet dining halls (Thiagarajah and Getty, 2013).

Determining the exact amount of food waste requires a careful prediction of the number of individuals receiving food services (Pirani and Thompson, 2016). This issue becomes even more serious when free food or subsidies are distributed, because when users enjoy a free lunch, it is likely for them to eat out somewhere else. Implementing a reservation system that compels users to book meals in advance may contribute to planning and production in the cooking sector in order to prevent overproduction (Steen et al., 2018). In a condition where demand for food service and uncertainty of the available information are high, efforts made to strengthen the reservation system proved to be significant (Ongkunaruk and Kessuvan, 2013).

Since the accurate prediction of demand has a significant role in reducing food waste (Pirani and Thompson, 2016), the present study consciously decided to focus on this issue. In the field of demand estimation with uncertainty in reservation systems, several studies have been conducted, mainly focusing on the fields of hotel management (Fiori and Foroni, 2020; Lacagnina and Provenzano, 2016) and transportation planning (Gao and Le, 2018; Luo and Shi, 2006); however, the present research takes a university reservation system as a case study and aims to accurately estimate the exact amount of meals needed based on the booking data and demand-side information. Although several studies have been conducted in the field of reservation systems (Tsai, 2014) and their relationship with food waste management (Ongkunaruk and Kessuvan, 2013), designing and implementing a demand estimation model under uncertainty in order to manage and reduce food waste in university self-services has received scant attention.

The case study of the present research was inspired by a policy of cheap food distribution through government subsidies. The final price of food turns out to be very low by such student food subsidies. This is the reason why students are not fully adhering to their reservations, which will lead to their no-show during servicing hours. Taking the no-show and over-show rates into consideration has a significant role in determining the amount of food waste (Pirani and Thompson, 2016). The present study focuses on food waste reduction, separating the reserved and actual demands. In order to take control of this issue, a model is proposed here that includes a two-step estimation using a machine learning model based on artificial neural networks (ANNs) to ensure significant accuracy of demand prediction. In step 1, the

Table 1. The amount of food waste due to reservations and the absence of students.

Year	Number of untaken meals	Rate of untaken meals (no-shows) (%)	Estimated weight of total waste (kg)
2013	141,066	18.70	59,248
2014	93,631	17.34	39,325
2015	87,020	15.61	36,548
2016	83,435	15.09	35,043
2017	86,417	14.13	36,295
2018	94,411	14.65	39,653
2019	121,906	14.68	51,201

definite demand component is estimated and, in the next step, that is, post-processing step, the uncertain demand component is estimated using error analysis.

The rest of this work is organized as follows: The next section discusses the case of a large university and investigates the obtained data and the proposed model. After that, section "Results and discussion" describes the results of implementing the proposed model based on the mentioned data and provides an appropriate response at the optimum level of confidence. In the final section, the output of the research is analyzed, hence bringing the benefits of this study to the fore.

Materials and methods

This section describes the major steps in the current study. First, an overview of the studied case is presented to clarify the aspects of the problem. Next, the structure of data and variables is identified. After this, the proposed machine learning approach for demand estimation in food waste reduction is demonstrated, followed by an uncertainty evaluation for cost minimization.

Study area

This study was conducted based on relevant information about one of the major universities in Tehran, Iran. The university has nearly 14,000 students, 4219 of whom live in dorms. It provides three meals for students: breakfasts, lunch, and dinner on week days (except holidays) that breakfast and dinner served only for dormitory students. Meals are distributed at different dining halls of the university, some of which are located on campus and some in dormitories.

Based on the usual procedure of reservation and distribution of food at this university, each student can choose his/her desired food out of possible options (based on food name, food price level, and place of receiving food) in a certain period of time before each mealtime and booking it online. The cost of this reservation will be paid out of a student's online wallet, which needs charging. At the end of the reservation, students will not be able to reserve a desired meal and will eventually be able to return to their pre-selected location at the due time to receive their food. Breakfast and dinner meals are provided only for those who live in dormitory and hence, have more stability of reservation and distribution due to delivery to dorms or breakfast serving as multiple packages once a week. Therefore, the number of booking transactions for breakfast or dinner is lower than that for lunch. Meanwhile, universities usually hold similar approaches and standards in terms of dining services. In order to develop a comprehensive and generalizable model, the present study investigated the lunch menu served at this university at the central dining hall (main self-service). The meal menu includes two optional foods that the student can book up to 48 hours prior to the due day. Usually, a significant percentage of students do not show up to receive the meal and their reserved food will remain unconsumed/untaken. This amount of unconsumed/untaken food is defined as food waste, according to the definition presented by Okazaki et al. (2008), which requires management.

According to the data collected during this study, from 2013 to 2019, the volume of food waste due to the absence (no-show) of students after lunch and dinner reservation is described in Table 1. An estimation based on the university analysis revealed that the average weight of each food portion is 420 g; the weight of food waste related to each year is shown separately in the third column of the Table 1.

Student food reservation and delivery systems are not active due to the summer holidays except August, and students are served for the rest of the year. A significant amount of food leftovers may be consumed through free distribution among predetermined charities or student dormitories. Further, two types of food waste respectively derived from unserved and served meals in the form of leftovers turn into university garbage. In other cases, such food leftovers may be sold to different students or other individuals for a different fee.

Data collection

The initial step in collecting data for demand prediction is determining the attributes or characteristics associated with the problem. To this end, the literature on demand forecasting in university dining systems was reviewed. However, few researchers have analyzed the characteristics of demand prediction and its many aspects in the field under study. Table 2 shows a number of demand prediction aspects explored in the previous corresponding research studies for estimating the demand rate.

To cover all the aspects given in Table 2, the current paper considers various attributes that influence students' demand.

Table 2. Aspects of demand prediction in dining systems.

Aspect	Researches
Menu or type of foods	Blecher and Yeh (2008); Kılıç (2015); Yergök et al. (2018); Çetinkaya and Erdal (2019); Yurtsever and Tecim (2020)
Day of week or month	Ryu et al. (2004); Blecher and Yeh (2008); Yergök et al. (2018); Ali et al. (2019); Çetinkaya and Erdal (2019); Yurtsever and Tecim (2020)
Specific times such as holidays	Kılıç (2015); Xinliang and Dandan (2017); Yergök et al. (2018); Çetinkaya and Erdal (2019); Yurtsever and Tecim (2020)
Individual factors	Kılıç (2015); Xinliang and Dandan (2017); Yurtsever and Tecim (2020)
Price or budget	Yurtsever and Tecim (2020)

Table 3. Details of the attributes used in estimating the show rate.

Attributes	Type	Columns
The day of week	zero/one	5
The day after/before a holiday	zero/one	2
Price level of Food 1/Food 2	zero/one	3+3
Name of Food 1/Food 2	zero/one	10+10
Total number of reservations for Food 1/Food 2	Real number	1+1
The rate of reservations for Food 1/Food 2 by undergraduate students	Between 0 & 1	1+1
The rate of reservations for Food 1/Food 2 by Masters' students	Between 0 & 1	1+1
The rate of reservations for Food 1/Food 2 by PhD students	Between 0 & 1	1+1
The rate of reservations for Food 1/Food 2 by dormitory students	Between 0 & 1	1+1
Ratio of previous show rate of reservoirs of Food 1/Food 2	Between 0 & 1	1+1

Extracted attributes are weekdays, consecutive holidays, food price level, food name, number of reservations segregated by academic degree, number of reservations segregated by accommodation status, and the ratios of previous attendance. Although previous studies have proposed several main aspects of demand, none of them performed the demand prediction under reservation data. Therefore, utilizing the mentioned novel attributes, this study considers booking parameters to complete the proposed model. Furthermore, individual factors are given significance by incorporating academic degree, accommodation status, and usual participation in dining halls. These attributes were extracted from the opinions of university experts, thanks to their former experience and knowledge of students' behavior.

They are the employees and administrators of the university's student nutrition department and have accumulated ample experience and knowledge of students' behavior in terms of meal booking and attendance at meal hours. Furthermore, it was attempted to use related experts in these positions at other universities. In addition, this study attempted to collect the views of information-technology experts who have worked extensively on this subject.

The data of this study, which include a total of 886 records, are related to lunch meals served at the university's central dining hall from 2013 to 2019.

The users' show rate is the target variable of this research. This variable represents the ratio of the number of users who have booked meals through the system and come to receive the service. From the two food options offered in the reservation system, each student can make a reservation for one of the recommended meals. Therefore, with the exception of such date indexes as weekdays and consecutive holidays, the rest of the attributes for each food have a separate, specific value. The description and characteristics of each attribute are listed separately in Table 3. The index of history is the converted history of each day, which is replaced by an integer. The day of week refers to the day number in a week that varies from 1 to 5 (in the form of five input variables with one hot encoding format), given that this service is not active on the weekends. In case of a holiday after/before a certain day, the index associated with that day returns to 1. The price of foodservice is constant and categorized into three levels of high/medium/low values. Therefore, food price level is converted into three variables of zero/one. Given that 10 different types of food are served by the university, 10 zero/one attributes are also considered as food names. The total number of reservations indicates the booked demand for the corresponding food. Each of the attributes including the rate of reservations by undergraduate/graduate/postgraduate students and rate of reservations by dormitory students to all students that booked meals is represented by a number in interval [0,1]. The number of dormitory students is considered given that those who live in dorms may treat the dining system differently because of their high participation rate on campus. Further to that, the main supply source of food for them is the university dining system. The last attribute for each food is the ratio of the previous show-up of users, which is a number between zero and one. This figure suggests that the users with prior reservations showed up to receive their food in person in only a percentage of cases on average.

Food waste reduction model

The main purpose of this study is to reduce food waste under uncertainty. There are many approaches that have been pursued to deal with uncertainty. Given that the current study tries to estimate the actual demand for reducing untaken meals, a perdition approach should be applied to estimate the demand. Therefore, selecting an appropriate prediction tool plays a critical role in this task. Artificial Neural Networks are among the most non-parametric methods for solving function estimation and fitness problems that are learned through a set of input/output data (Zhang and Suganthan, 2016). An important advantage of neural networks is the ability to learn complex nonlinear systems (Abbasi and El Hanandeh, 2016), thus making them a good option for challenging issues like the one dealt with in this study. Although neural networks in many cases outperform traditional models of the estimation function with respect to demand estimation (Muriana, 2017), the application of neural networks to predict outcomes under uncertainty or in case of constant systemic changes is quite challenging (Khosravi et al., 2011). This study considers the aforementioned concern. Due to the uncertainty in and instability of user behavior after reservation (Painter et al., 2016), the usual implementation of neural networks to ensure point predictions is not enough. Therefore, post-processing methods are appropriate options to modify and reduce errors (Jorgensen and Linneberg, 1999; Siswantoro et al., 2016).

The proposed model is based on two steps: in the first stage, a point prediction is carried out; in the post-processing stage, the desired point is reached by simply analyzing the expected level of confidence based on the minimum possible cost. Having applied this approach in their study, Vergara-Dietrich et al. (2019) aimed to manage renewable power systems effectively. They combined ANNs based on time series with chance-constrained programming. The considered variable is modeled as a variable with definite and random components; the definite component was estimated by the neural network and processed along with its random counterpart in the desired constraint. The proposed model in this study is similar to the recent compound, except that in addition to the estimation of definite component, the random component is also estimated as a cumulative distribution function during the learning of ANNs. The steps and details of the ANN-based model are described below.

The framework for estimating the presence of users. In the proposed model of the present research, the amount of actual demand is expressed in terms of the show rate. According to equation (1), the show rate r represents the percentage of the booked demands served at the mealtime:

$$\tilde{r} = \frac{\tilde{d}}{D} \tag{1}$$

where $\tilde{d}$ and D denote the number of users who succeed to arrive at the designated time (which is an unknown value) and the number of booked demands, respectively. Due to uncertainty in the value of the random demand $\tilde{d}$, the value of $\tilde{r}$ is also subject to uncertainty.

Herein, the objective of the proposed model is to determine the number of requirements for food production or cooking. Considering the university's commitments to students who have made service reservations, decision-makers should make relevant decisions under uncertainty at a sufficient level of confidence. If x indicates the amount of the required food, relation 2 shows its relation to the random demand $\tilde{d}$ at a desired level of confidence $(1-\alpha)$:

$$\Pr\left(x \geq \tilde{d}\right) \geq 1-\alpha \tag{2}$$

Based on equation (1), since the parameter $\tilde{d}$ is a function of the user's show rate, relation 2 can be substituted as bellow:

$$\Pr\left(x \geq \tilde{r}.D\right) \geq 1-\alpha \tag{3}$$

The probabilistic relation 3 is actually unknown because distribution of daily show rate is not accessible. Although the estimated distribution obtained by the ANN can be considered instead of the distribution $\tilde{r}$, the error resulting from neural network estimation can cause a significant deviation. Therefore, estimation by the neural network and the error resulting from the random behavior that cannot be predicted by the neural network can be used to assess the random parameter. This approach is expressed in:

$$\tilde{r} = \hat{r} + \tilde{e} \tag{4}$$

where $\hat{r}$ (definite component) is an estimation by the neural network and $\tilde{e}$ (random component) is error indicator. Therefore, the inequation (3) of the model can be rewritten as:

$$\Pr\left(x \geq \left(\hat{r} + \tilde{e}\right).D\right) \geq 1-\alpha \tag{5}$$

Considering the function obtained from the cumulative distribution fit $\tilde{e}$, relation 5 can be substituted into inequation (6):

$$\mathrm{F}_{\tilde{e}}\left(\frac{x}{D} - \hat{r}\right) \geq 1-\alpha \tag{6}$$

Artificial neural networks-based model

Preliminaries. By considering the nature of fitness and obtaining $\tilde{r}$ through ANNs, the default structure of neural network in this research is shallow based on feedforward framework with a single hidden layer. Assuming that T is the number of training samples in $\left\{\left(\mathbf{p_t}, \mathbf{r_t}\right), t = 1,\ldots,T\right\}$ with $\mathbf{p_t} = \left(p_{t1}, p_{t2},\ldots, p_{tn}\right) \in \mathbb{R}^n$ and $\mathbf{r_t} = \left(r_{t1}, r_{t2},\ldots, r_{tm}\right) \in \mathbb{R}^m$ denoting respectively the input vectors and the targets in a supervised learning scheme, a feedforward neural network with N nodes in the hidden layer and transfer function ϕ is represented as follows:

$$Y_N(\mathbf{p_t}) = \sum_{i=1}^{N} \boldsymbol{\beta}_i \phi\left(\mathbf{w_i}.\mathbf{p_t} + b_i\right),\ t = 1,\ldots,T \tag{7}$$

in which equation (7), $\mathbf{w_i} = \left(w_{i1}, w_{i2},\ldots, w_{in}\right)^T$ is the vector of input weights that binds the input nodes and the i th hidden

node. $\boldsymbol{\beta}_{\mathbf{i}} = (\beta_{i1}, \beta_{i2}, \ldots, \beta_{im})^T$ is the output weight vector that binds the i th hidden node and the output nodes. b_i is the bias of the i th hidden node. Therefore, achieving the least error in estimating is equivalent to finding the values of b_i, $\boldsymbol{\beta}_{\mathbf{i}}$, $\mathbf{w}_{\mathbf{i}}$ that minimizes function 8 (Lolli et al., 2017; Siswantoro et al., 2016).

$$\left\| Y_N(\mathbf{p}_t) - \mathbf{r}_t \right\|, t = 1, \ldots, T \tag{8}$$

Two-step estimation based on ANNs. After determining the value of $\tilde{r}$ based on the results of the mentioned network, equation (4) can be used to obtain the value of $\tilde{e}$. The second step is to apply second neural network to reduce prediction errors of the probabilistic distribution function $\tilde{e}$. This assessment has been made using the ANN according to equations (7) and (8), except that the inputs and the output are prediction errors in the first stage (e) and empirical cumulative distribution function of e, respectively.

Determining the α level based on the lowest cost analysis. The results of the proposed model are strongly dependent on the amount of α in relation 6. The higher this value is, the larger the solution space of the inequation and the smaller the x value will be. For this reason, the performance of the proposed model is evaluated based on not only the volume of food waste or its material cost, but also the impact of lack of food needed when students show up to get the meal under food shortage.

Therefore, this research focuses on the correct estimation of the amount of actual demand to significantly reduce the number of untaken meals. Based on the current state of service, any reduction in the amount of untaken food is equal to a considerable reduction in cooking. In other words, if no food is distributed, the highest level of the desired point is achieved. In practice, however, there has to be a trade-off between the cost of food waste (or the cost of surplus food with no demand for it at the mealtime) and the cost of food shortages; both of these costs are directly dependent on the decision-making about the amount of food production. These two costs are shown in equation (9) as C_w and C_s, respectively.

$$\mathrm{C}(x) = \mathrm{C}_w(x) + \mathrm{C}_s(x) \tag{9}$$

According to the reservation management literature which is rooted in revenue management, the shortage cost is actually the denied-service cost. According to a realistic assumption made by Talluri and Van Ryzin (2004), the penalty for shortage in the reservation management models should be in the increasing and convex form which will include the cost of an alternative service, compensation, and loss of goodwill (He, 2019). Given the specific nature of the proposed case study which is arising from government subsidy, the structure of the costs of unserved services in a competitive environment varies. However, one cannot ignore the ascending pattern of shortage cost, because if the proposed model of this study under-predicts the outcome by a margin of one person (from the actual number of actual demands) as compared to the margin of dozens of students, the costs incurred by the university or the disgrace ascribed to quality of university services will significantly vary. In this regard, Sillanpää et al. (2020) implemented the shortage cost function in the form of a piecewise linear function to increase the shortage cost at higher levels of divergence between demand and serviceable quantity. Similarly, in this study, the piecewise linear function was taken into consideration, except that a fixed threshold limit was used. Therefore, the function of shortage cost is given as follows:

$$\mathrm{C}_s(x) = c_{s_v}.(t.D - x) + c_{s_f}.\mathrm{sgn}(t.D - x - \delta) \tag{10}$$

where the cost parameter c_{s_v} represents the rate of service compensation for each unmet demand. δ indicates the threshold of increase in the shortage cost; in other words, if the actual value of demand exceeds the predicted value, separately calculated c_{s_v} as well as the constant cost parameter c_{s_f} will be considered as the penalty costs to compensate the unmet demands of student associations, loss of mutual goodwill, and legality and credibility issues derived from the accumulated volume of unsatisfied demands.

Accordingly, the considered cost model according to equation (10) turns is substituted into equation (11):

$$\mathrm{C}(x) = c_w.(-t.D + x)^+ + c_{s_v}.(t.D - x)^+ + c_{s_f}.\mathrm{sgn}(t.D - x - \delta)^+ \tag{11}$$

where cost parameter c_w indicates waste cost due to the greater amount of the predicted demand as compared to the actual amount of demand per case.

Based on the details mentioned in this section, sequential stages of the proposed model are shown in Figure 1. The stages of the proposed model are shown in this section.

Results and discussion

The data used in this research, which are based on the attributes in Table 3, were extracted from the university dataset under study. These data are applicable to the analysis model of the research only after being pre-processed (including the elimination of the noises, days without food or with incomplete data, and integration of records based on the two desired outputs). To estimate the value of $\tilde{r}$ in equation (4) using neural network, the acquired data were partitioned into the three following parts based on a common ratio (Pandey et al., 2016) to prevent overfitting: training, validation, and testing at ratios of 70%, 15%, and 15%, respectively. To achieve the best ANN structure for estimating $\tilde{r}$, different structures were examined. These structures include a combination neural network's parameters in a permutational analysis. Two training functions (Levenberg-Marquardt and Scaled Conjugate Gradient), a range of 10–60 neurons for the hidden layer and two transfer functions (Log-sigmoid and Hyperbolic Tangent Sigmoid function) was considered. The best architecture was evaluated based on the mean squared error, mean absolute error, and coefficient of determination (Abbasi and El Hanandeh, 2016; Pandey et al., 2016) on test part of data. Therefore, an ANN with 10 neurons in a hidden layer along with the Log-sigmoid transfer function in the

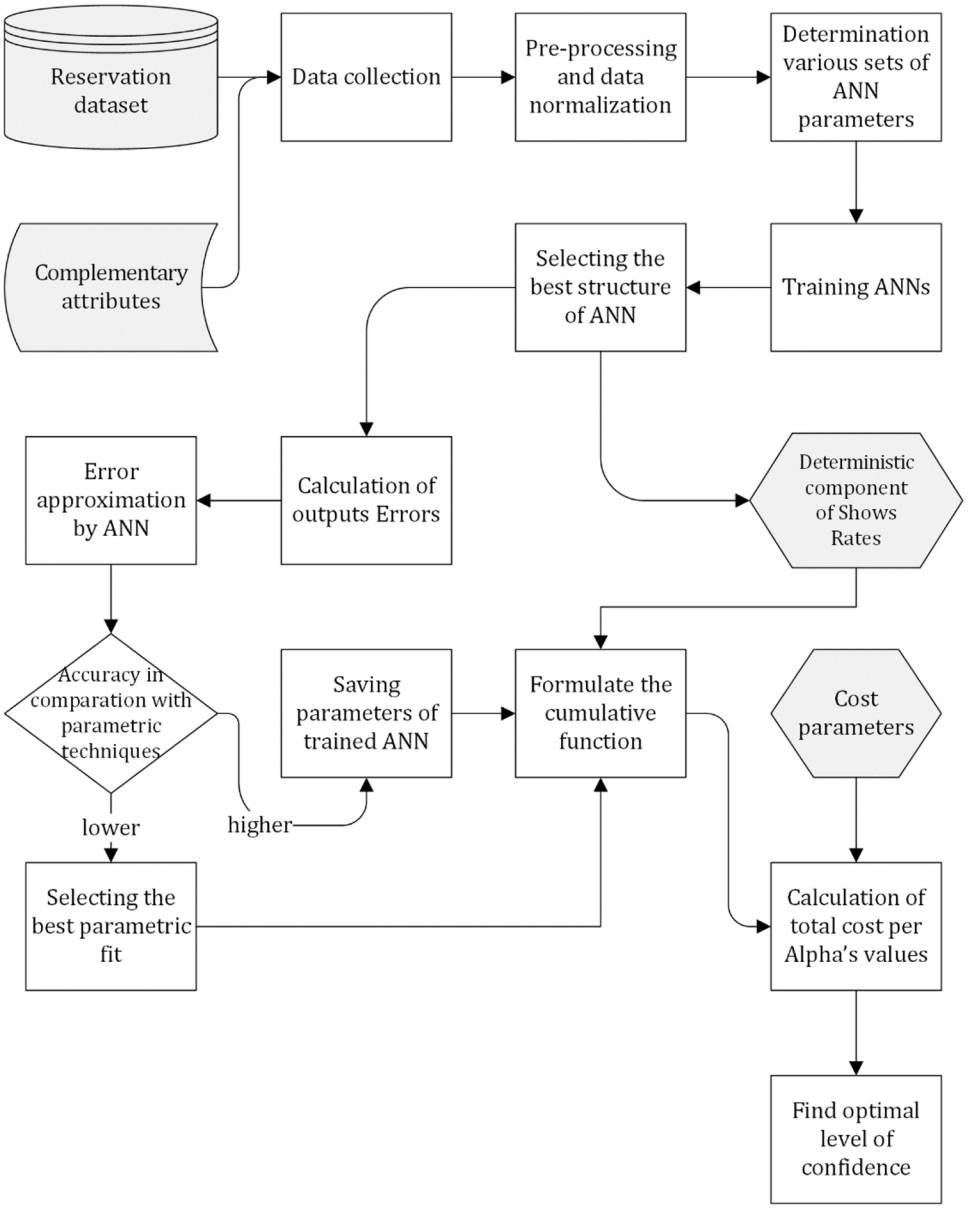


Figure 1. The flow of the model's stages.

backpropagation process with the Levenberg-Marquardt training algorithm was selected as the best structure. The performance results of trained neural networks based on the regression between the actual value and the estimated value with the accuracy rates of 73%–75% are shown in Figure 2.

According to equation (4), the probability distribution function of $\tilde{e}$ is used to estimate the random component. Accordingly, using the difference between the values of the actual show rate and the output of the trained neural network, the ratio of the error values to the target value is calculated. Figure 3 shows the histogram of the distribution of error values and the best distribution fitting each of them.

By evaluating different probability distribution functions on the data relevant to the neural network error in both food predictions, the appropriate probability distribution has been fitted to each of these data. The distribution of the first data is Epsilon Skew Normal with the parameters $\theta = 0.0116$, $\sigma = 0.0231$ and $\epsilon = 0.2378$. The proper distribution of the errors of the second data follows t-Location-Scale with the parameters $\mu = 0.0032$, $\sigma = 0.0171$, and $\nu = 5.3720$. Nevertheless, compared to these two distributions, the estimation of the cumulative distribution function of the error values was examined using another ANN. The output of this fitness compared with each of the two distributions in the corresponding fitting is shown in Figure 4. For evaluation of the neural network in comparison with the classic distribution functions, equation (12) is used (Chen et al., 2020). This performance metric is known as the *mean absolute percentage errors* (MAPEs).

$$M_f = \sum_{k=1}^{m} \frac{\left|F_k - \tilde{F}_k\right|}{F_k} \times \frac{100\%}{m} \tag{12}$$

The MAPEs of the estimated CDF for the neural network are equal to 1.38 and 1.34, while they are equal to 5.57 in Epsilon

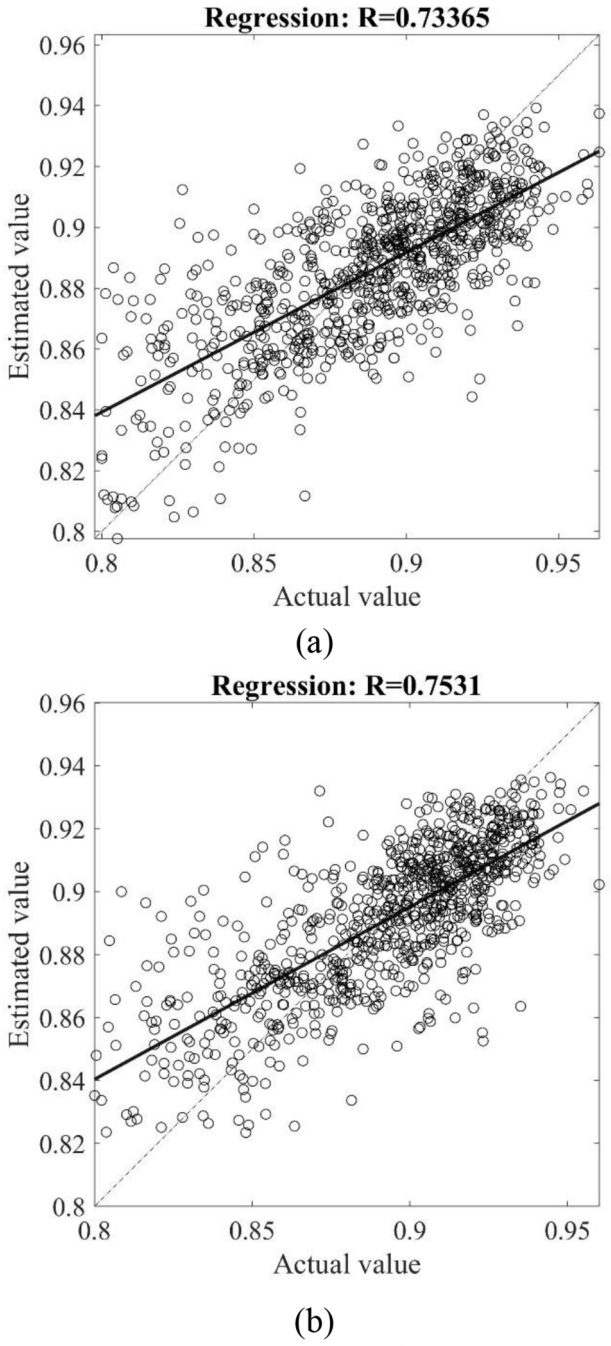


Figure 2. Correlation between neural network estimation and actual show rate: (a) Food 1 and (b) Food 2.

Skew Normal distribution fitness on the first food and to 6.39 in t-Location-Scale fitness on the second food. Accordingly, the application of neural networks ensured a favorable performance in estimating the cumulative distribution function. In addition to their high performance, the convenience and simplicity of applying the mentioned function due to neural networks in 8 is another benefit, because no new probabilistic distribution function is replaced and the mathematical structure of constraints in the function does not change due to user behavior variations over time. In the structure of neural networks, a sigmoid symmetric function is used as a transfer function in the hidden layer;

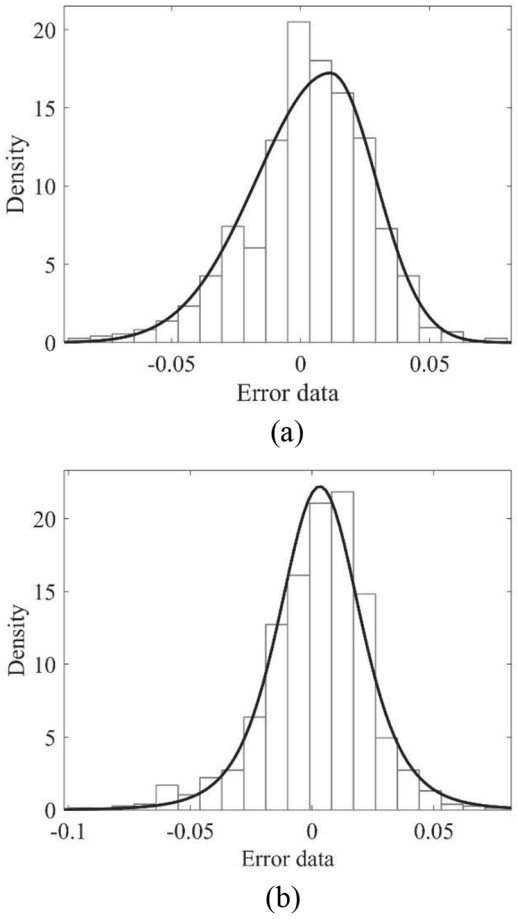


Figure 3. Histogram of the output errors of neural network in estimating the show rate accompanied by fitting of the appropriate distribution function: (a) Food 1 and (b) Food 2.

therefore, the function derived from network learning can be written in equation (13) as follows:

$$E_x = \mathrm{SL}\left[b_2 + \mathbf{w}_2^T \times \left(\frac{2}{1+\exp\left(-2\left(\mathbf{b}_1 + \mathbf{w}_1 x\right)\right)} \right) \right] \qquad (13)$$

where $\mathbf{b}_1$ and b_2 are the biases of input and hidden layers, respectively, and $\mathbf{w}_1$ and $\mathbf{w}_2$ are the weight vectors of input layer to hidden layer and hidden layer to output layer, respectively. SL function is in fact the transfer function of output layer that restricts the output value between 0 and 1 in a linear manner. As a result, network estimates in the form of cumulative distribution function work properly, as expected.

By setting cost parameters c_w, c_{s_o} and c_{s_j} to 120,000 IRR, 360,000 IRR and 10,000,000 IRR, the proposed model was evaluated with different values of α from 0.01 to 0.3 according to equation (11). The performance of the model was evaluated at a time interval of 1 year (2019), as shown in Figure 5, and the threshold limit δ is equal to 10 people. The values of food waste cost and shortage penalty are set based on the minimum value of x.

The minimum total cost is obtained in $\alpha = 0.22$ and $\alpha = 0.19$, respectively for Food 1 and Food 2. If the proposed final model

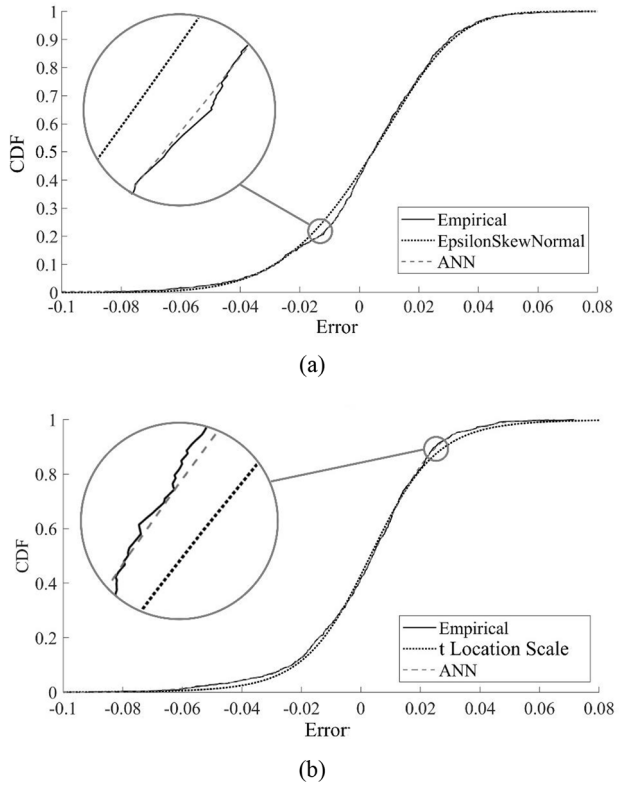


Figure 4. Comparison of CDF estimation by ANN and the distribution function candidates on the data derived from prediction error: (a) general view of estimated CDF for errors of Food 1 and (b) general view of estimated CDF for errors of Food 2.

is run with the obtained values of α , the acquired output will be consistent with the value in Figure 6. The reduction rate of food waste in 2019 is shown in this figure. In addition, Figure 6 shows the estimated reduceable rate of food waste by the proposed model. Meanwhile, based on analysis of the total cost for 2019, this Figure shows the total cost estimated by the proposed model with the data received in 2019.

According to Figure 6, waste costs were obviously reduced due to a significant reduction in the amount of food waste. Moreover, this decrease is in equilibrium and consistent with the shortage cost, which has finally resulted in the total cost reduction. Although shortage costs are added to the new model, the total cost in 2019 is still lower than the total cost in the current situation. Although the monthly amount of waste determined by the new model is lower than the current waste in each month, the trend of the total cost in February goes against that in other months. The main reason for such higher costs lies in the nature of February as it is located between two semesters of the academic year and it includes exam days. Therefore, the behavior of students in this period of time is subject to changes. The total reservation rate is lower than that in regular time periods, leading to changes in the data pattern. It can be concluded that the proposed model has underestimated the actual demand in comparison to demands in usual time periods, thus imposing high penalty costs for unsatisfied demand. However, the total cost, as

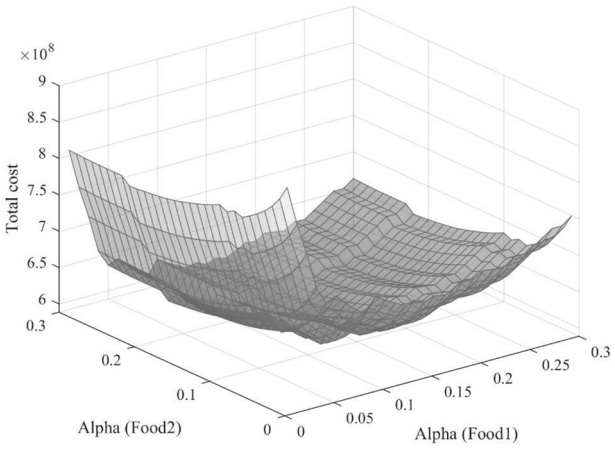


Figure 5. Total costs including waste cost and shortage penalty based on different values of α.

an annual aspect, is lower than current cost; hence, the model has correctly reduced the total annual cost.

A significant reduction in the amount of food waste was observed in different months of 2019, except for August when the university system is usually inactive. This volume of waste reduction varies from 63.36% to 99.28% per month and, in general, a 79.66% reduction in food waste was recorded throughout the year. This result has been obtained from the difference between the current untaken meals and the estimated untaken meals generated by the proposed demand prediction. Therefore, the amount of reduction was obtained by the ratio of proposed model-based estimated untaken meals over reported untaken meals in 2019. The reduction rate of waste would be determined if the new model was established for the year 2019. However, according to the data listed in Table 1 in 2019, 14.68% of users' reservations generated food waste due to their absence (no-show). If the current model is successfully generalized to the entire university while keeping the response quality, the amount of food waste will be 10,414 kg in 1 year. Such a reduction in food waste works in conjunction with controlling the risk of improper food services, because determining the costs of shortage and waste is still controversial and up for debate.

By keeping in mind, the negotiated, final costs of food for the university, the waste cost can be easily determined. However, determining the cost is no longer easy in case of shortage. In addition to the direct cost of food replacement, compensation costs and damaged goodwill (He, 2019) should also be considered in the final tally. Determining these two costs, that is, compensation and damaged goodwill costs, becomes more challenging at the end of each service day. The queue is always governed by the *first-in first-out* rule, hence the reason if some students do not receive their meals, it means they would be among the last ones to stand in the food delivery queue. Therefore, food shortage often occurs in the final minutes of mealtime.

According to the average number of users' show-up from a certain time to the end of the mealtime, if the reserved food runs out

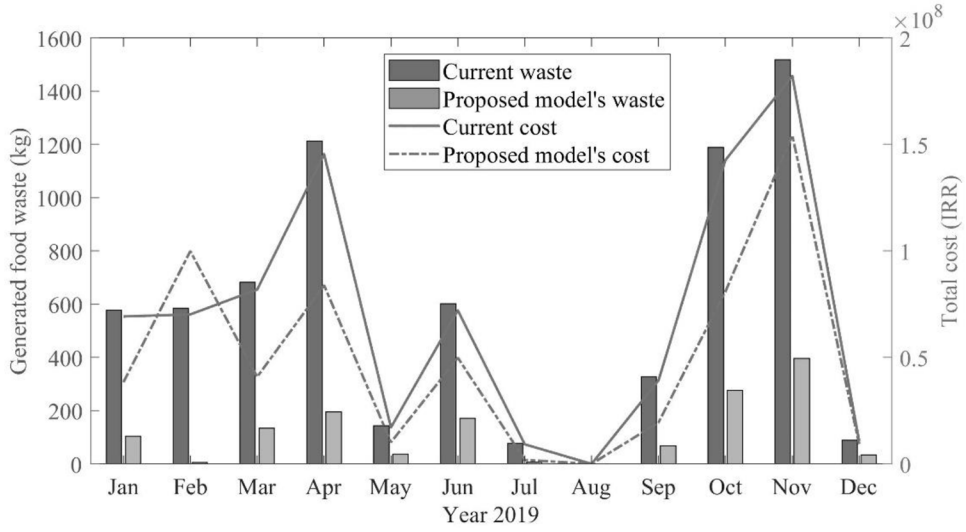


Figure 6. Estimated cost and waste reduction during 2019.

within the last 15 minutes of the servicing process, an average of 25 users may not receive their intended services. Therefore, if the volume of food is not enough, there will be a very little chance to compensate for the delay in serving the students, because preparing food, even through other suppliers, requires more time and money for which the student may not be able to wait more. Therefore, the parameters of this section are determined according to the university policy-making and its commitment to promoting the stability of servicing the students in university dining halls.

Conclusions

Waste management at universities has received much attention in the field of waste management and led to further research. Most of the related papers in this field focused on the procedures and factors involved in waste production, analysis of waste volume, and management of recycling programs and waste reduction policies (Gallardo et al., 2016). Although many studies have considered food waste in universities, no comprehensive studies have been conducted on demand prediction based on reservation data. There is a research gap in the context of food waste management of dining systems that include meal booking and food distribution. Therefore, one of the main contributions of the present study is filling the mentioned gap in analysis of university food waste. In addition to the aspects explored in previous studies for demand prediction, the characteristics of meal booking data and reservation segments has been considered in the proposed model. The present research represents a centralized effort to predict the proper volume of cooking so as to significantly reduce untaken meals. The resulting wastes are generally categorized as PCFWs that the accurate prediction of the amount of food consumption or actual demand at the universities' dining halls will have a significant impact on controlling food waste. The novelty of the present study lies in analyzing the presence and absence (show and no-show rates) of students at the dining halls and presenting a related mathematical model, likes of which have not been seen in the previous research studies. Although previous researchers have analyzed the reasons for students' absence in self-services to receive their reserved meals (Painter et al., 2016), only a few studies have provided a model based on quantitative estimates of the students' behavior under random situations with different uncertain costs.

Based on investigating the behavior of students in response to their reserved meals, a set of random or uncontrolled factors including students' drowsiness or satiation and crowded self-services was given and analyzed (Painter et al., 2016). Therefore, a careful prediction of the effects of these factors, which vary from individual to individual, as well as an initial estimation, requires sensitivity analysis and confidence level assessment in random conditions. In this regard, the first phase of the present study provided an estimation obtained from the ANN, while the second phase, that is, the combination of neural network results based on the random behavior of the error generator, made it possible to consider neglected random incidents or behaviors of model attributes in the form of a probabilistic inequation in the actual demand estimation model. To this end, the proposed model carried out the corresponding estimations in accordance with fluctuations of random behaviors, the results of which were discussed in the former section. The two-phase model with analysis of uncertainty led to a 79% decrease in the food waste of the considered university. The reduction was based on decreasing the number of untaken meals that can reduce food waste directly. To implement this approach, a simultaneous balance between minimizing waste cost and shortage penalty was considered. Therefore, the reduction in food waste volume has been considered together with controlling the shortage and waste cost. Furthermore, emphasizing the significant reduction in food waste by the proposed framework, novel attributes utilized in prediction part of the model can be approved and used in future establishment of demand prediction at universities with booking data.

Furthermore, the proposed model managed to create a notable potential for defining and implementing other executive rules and service policies of the mentioned university. Since the proposed problem is not just a model limited to fitting and is defined in the form of a probabilistic inequation, it will be capable of designing

and solving larger complexities common at any university. This capability can be generalized through the chance-constrained programming model. The generalization of such possible planning could help decision-makers take into account other constraints and limitations involved. Choosing two types of food, each as an alternative or a supplement to the other, is another aspect that can promote the model efficiency. Although students have already set their preferences during the reservation, they may still face shortage; in this case, it may be possible to replace food 1 with food 2 or vice versa at a new price. This idea would realize the objective of food waste reduction.

This study has been subject to certain limitations from the very beginning. Therefore, to extend this research and provide suggestions for future research, the data of this research should be injected into two datasets to which the existing authors did not have access so that the impact and capability of the proposed model would be promoted. During the course of conducting this study, the data on the daily curriculum of students were not made available to researchers. Therefore, among the attributes that could not be examined are the daily schedule of users and the hours before and after serving food at the university's self-services. In addition, the data record associated with the behavior of users after reserving their intended food was not sufficiently stored in the database of the university reservation system. In case these data become available, behaviors such as canceling food reservations after the authorized interval (in the form of selling food to other students) can also be considered to be added to the model. In case the aforementioned behavioral patterns are seen on a certain day, the possibility of a new emerging behavioral pattern is reinforced.

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